Kai-Hua (Kelvin) Wang

(412) 209-9357 | kaihuawang.work@gmail.com | Personal Website

EDUCATION

Carnegie Mellon University (CMU) *Pittsburgh, PA*
M.S. in Mechanical Engineering Expected: May 2028

- Focus Area: Robotics, Optimization, Autonomous Systems, and Controls

National Taiwan University (NTU) *Taipei, Taiwan*
B.S. in Mechanical Engineering Sep 2021 – Jan 2026

- Overall GPA: 3.79 / 4.3, Last 60 credits: 4.12 / 4.3

TECHNICAL SKILLS

Programming Languages	C++, Python, C, MATLAB
Robotics & Controls	SLAM, Vehicle Dynamics, State Estimation, Controls, Sensor Fusion
Frameworks & Libraries	ROS/ROS2, PyTorch, OpenCV
Engineering Tools	Simulink, SolidWorks, COMSOL, STAR-CCM+
Languages	English (Proficient), Mandarin (Native)

WORK EXPERIENCE

Garmin Ltd. *New Taipei City, Taiwan*
Manufacturing Engineer Intern, New Product Introduction (NPI) Jul 2024 – Jan 2025

- Calibrated automated inductive-button test equipment and validated test performance with electrical engineers, reducing the retest rate by 17.6%.
- Led root-cause analysis and refined GPSMAP H1 fixtures to prevent component displacement and damage.
- Developed assembly workflows and fixture specifications using motion-time analysis and PFMEA; authored standardized work instructions to support the transition from NPI to mass production.

RESEARCH EXPERIENCE

Trustworthy Robotics, Intelligence, and Optimization Lab, CMU; Advisor: Frederike Dümbgen *Aug 2026 – Present*
Robotic Pushing: Learning-Based Action Optimization

- Developing context-aware action-selection methods for planar robotic pushing, using predicted task costs conditioned on object, goal, and obstacle configurations to guide robot actions.
- Building simulation benchmarks with Model Predictive Path Integral(MPPI) control to evaluate goal-reaching and obstacle-avoidance performance and generate rollout data for task-cost modeling.

System Optimization Laboratory, NTU; Advisor: Kuei-Yuan Chan *Sep 2024 – Jan 2026*
Autonomous Straddle Carrier: Redundant Localization and Safety

- Designed and built a 1:12-scale autonomous straddle carrier with a motorized hoist, integrating mechanical, electrical, and ROS2 software subsystems on a Jetson Orin NX.
- Integrated LiDAR SLAM and AprilTag-based localization into the ROS2 navigation stack.
- Implemented a safety architecture combining LiDAR protective fields, a mechanical bumper, and an independent hardware emergency-stop circuit.

Autonomous Driving: Perception and Control (Foxtron)

- Led a seven-member team to develop a low-speed autonomous driving system for a Luxgen N7 EV, integrating YOLOv8 cone detection, RealSense D455 depth sensing, and Pure Pursuit path tracking.
- Developed a 14-DOF vehicle dynamics model in Simulink and tested perception–control integration in CARLA, achieving on-vehicle autonomous cone-lane navigation at approximately 10 km/h.

Smart and Advanced Manufacturing Laboratory, NTU; Advisor: Dian-Ru Li *Oct 2022 – Jan 2026*
Hemodialysis Catheter Stabilizer: Design and Evaluation (with Taipei Medical University)

- Designed and evaluated a hemodialysis catheter stabilization device in a 10-physician simulator study, achieving 38% shorter replacement time and 64% lower simulated blood loss versus baseline.

Formula SAE competition, Head of Aerodynamics, NTU Racing *Jul 2022 – Jul 2023*

- Led a seven-person team to design and validate aerodynamic packages using COMSOL and MATLAB, generating 229.8 N of downforce at 11 m/s and contributing to a first-place overall finish at Formula Student Taiwan.
- Introduced Resin Transfer Molding and optimized composite manufacturing, reducing aerodynamic-package weight by more than 25% while maintaining structural performance.